

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained **XCTest**-based automated test suites in **Swift** for core **Apple platforms**, ensuring robust application quality across devices.
- Implemented **LLM-powered test generation** capabilities, analyzing defect patterns to intelligently design new test cases and reduce manual effort by approximately **22%**.
- Designed and executed comprehensive automated testing strategies, integrating with **CI/CD pipelines** to enable continuous deployment and real-time reporting of test results.
- Contributed to the fine-tuning of **machine learning models** for intelligent defect analysis, improving the accuracy of automated bug reporting by **18%**.
- Collaborated with cross-functional teams to define and implement comprehensive test plans, focusing on **localization and internationalization** aspects of software releases.
- Analyzed detailed performance metrics from automated test runs, identifying bottlenecks and areas for optimization in application responsiveness and stability.

SKILLS

Programming Languages – Swift, Python, Java, SQL, R

Testing Frameworks & Tools – XCTest, GenAI, Large Language Models (LLMs), Machine Learning Model Training, Debugging

Platforms & Development – iOS, macOS, Apple Platforms, Application Development, CI/CD Pipelines, Jenkins, Git

Data & Analytics – Data Warehousing, Data Modelling, Power BI, Tableau, DAX, Power Query, Pandas, NumPy

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux

Enterprise Systems – ERP Integration Support, Payments Operations Dashboard, Transaction Monitoring, Reconciliation Workflows

PROJECTS

Automated Test Suite for Cross-Platform Application

- Engineered a comprehensive automated test suite using **Swift** and **XCTest** for a simulated enterprise application, covering UI interactions and API endpoints across **iOS** and **macOS**.
- Integrated the test suite into a **Jenkins CI/CD pipeline**, automating test execution on every code commit and providing immediate feedback on build stability.
- Developed custom reporting mechanisms to capture detailed performance metrics, identifying an average of **35 critical bugs** per release cycle that were previously missed.

LLM-Enhanced Test Case Generation and Analysis

- Developed a proof-of-concept using **Large Language Models (LLMs)** to analyze existing defect reports and generate novel test cases, increasing test coverage by an estimated **15%**.
- Implemented a system to fine-tune an **ML model** on historical bug data, improving the accuracy of predicting potential failure points in new feature releases.
- Designed and presented findings on the LLM's effectiveness to an internal engineering review board, demonstrating potential for significant efficiency gains in quality assurance.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)